

Probability - Homework 1

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1.2-17

(a)

To draw 4 cards with the same face value, there's 13 possibilities, then draw one from the rest of cards, there's $52 - 4 = 48$ possibilities (By definition, it's impossible to have same value with the previous 4 cards.), Hence, the probability is:

$$\frac{13 \times 48}{\binom{52}{4}} = \frac{1}{4165}$$

(b)

We can choose two type of from 13 face values, and for each face value we have 4 colors to choose, then we can choose either 2 or 3 cards from them (but not for both types). (btw, we should time a extra 2, for instance, if we need to construct a full-house using 2 and 3, we can draw 2 cards from face value 2 and 3 cards from face value 3, or draw 3 cards from face value 2 and 2 cards from face value 3, there's two possibilities.) Thus, the probability is:

$$\frac{\binom{13}{2} \binom{4}{2} \binom{4}{3} \times 2}{\binom{52}{5}} = \frac{6}{4165}$$

(c)

Choose 3 ranks from 13 ($\binom{13}{3}$), then decide which of the 3 ranks forms the triple ($\times 3$, since any of the 3 chosen ranks can be assigned the triple role), choose 3 suits for it ($\binom{4}{3}$), and choose 1 suit for each of the remaining 2 kicker ranks ($\binom{4}{1}^2$). Thus, the probability is:

$$\frac{\binom{13}{3} \binom{4}{3} (\binom{4}{1})^2 \times 3}{\binom{52}{5}} = \frac{88}{4165}$$

(d)

Choose 3 ranks from 13 ($\binom{13}{3}$), then decide which of the 3 ranks is the single kicker ($\times 3$, since any of the 3 chosen ranks can be assigned the kicker role), choose 1 suit for it ($\binom{4}{1}$), and choose 2 suits for each of the two pairs ($\binom{4}{2}^2$). Thus, the probability is:

$$\frac{\binom{13}{3} \binom{4}{1} (\binom{4}{2})^2 \times 3}{\binom{52}{5}} = \frac{198}{4165}$$

(e)

Choose 4 ranks from 13 ($\binom{13}{4}$), then decide which of the 4 ranks forms the pair ($\times 4$, since any of the 4 chosen ranks can be assigned the pair role), choose 2 suits for it ($\binom{4}{2}$), and choose 1 suit for each of the 3 remaining kicker ranks ($\binom{4}{1}^3$). Thus, the probability is:

$$\frac{\binom{13}{4} \binom{4}{2} \left(\binom{4}{1}\right)^3 \times 4}{\binom{52}{5}} = \frac{352}{833}$$

1.3-13

(a)

(b)

(c)

(d)

(e)

1.4-15

(a)

(b)

(c)

1.5-4